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KEYSTONE: Readiness Audits for Dispute-Key Availability in Encrypted Rollups

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Abstract

Encrypted optimistic rollups can keep ciphertexts and execution receipts available and still fail a dispute because the threshold service that must decrypt the disputed record is unavailable, late or uncooperative. We call the missing property dispute-key availability (DKA) and separate it from ciphertext availability, valid-share registration, audit-time readiness, authorized decryptability and deadline liveness. KEYSTONE audits readiness with sampled canary shares carrying Chaum–Pedersen proofs; their transcripts are simulatable from public shares, so audits exercise real shares without revealing any record. We give exact operating characteristics and show that a single small audit cannot separate a committee one share short of its threshold from a live one, whereas sequential tests can. Under a stated assumption, record-labeled TDH2 restores the confidentiality plain threshold ElGamal loses to parties obtaining other releases. For custodians that fail and recover, a hidden-Markov recursion gives the exact risk that a dispute fails after a given audit history, and a floor shows when auditing more often cannot help. Exact analysis, simulation and a 32-process testbed confirm these results, show that selective withholders pass every audit yet defeat every dispute, and, in a pre-registered replication, support an order-statistic deadline except without network delay.

Keywords

Dispute-key availability, threshold decryption, encrypted rollups, readiness audit, acceptance sampling, selective withholding

1 Introduction

Optimistic rollups execute transactions off chain and accept the result unless a challenger proves it wrong within a dispute window[1]. When the executed workload is private, as in encrypted machine-learning inference[2,3], the records a challenger needs are published only as ciphertexts. Data-availability mechanisms can keep these ciphertexts retrievable. They cannot ensure that the committee holding shares of the decryption key delivers enough valid decryption contributions once a dispute is authorized and before its deadline. A ciphertext that cannot be opened in time is, for the dispute, unavailable. AI-rollup disputes are our running example, but the question arises wherever a protocol must open threshold-encrypted inputs on demand.

The failure is silent until it is needed. Custodians may lose shares, fail together when a provider or region goes down, or run broken software, and the key must stay serviceable for a dispute window that lasts days and may outlive the committee that sealed the record. A strategic custodian may also answer every routine check and refuse one targeted dispute; no audit of the kind studied here can detect that, and we treat it as a limit. Threshold encryption, verifiable and proactive secret sharing, and proofs of retrievability and data-availability sampling [4–9] answer adjacent questions; none produces public evidence that enough custodians can serve an authorized dispute now, or states what such evidence implies.

We study this property, *dispute-key availability* (DKA), and design and evaluate KEYSTONE, a readiness audit for it that targets non-strategic faults. KEYSTONE seals each record under a fresh key wrapped by a threshold Diffie–Hellman KEM. Routine audits ask a beacon-sampled subset of custodians to apply their shares to a *canary*, a group element hashed from public audit context, and to prove correctness with a Chaum–Pedersen equality-of-discrete-logarithm (DLEQ) proof; each response is structurally a threshold-coin or threshold-signature share on a hashed message [10,11]. An authorized dispute applies the same share operation to the disputed record. Our contributions are:

1. **A property decomposition** (Section 4) of (ε, Δ_d)-DKA into ciphertext availability, valid-share registration, audit-time readiness, authorized decryptability, deadline liveness and pre-authorization confidentiality, with witnesses for three inferences a deployment must not draw.
2. **A confidentiality-preserving audit** (Section 5) whose canary transcripts are simulatable from public shares, so audits exercise real shares without opening records. The plain threshold KEM is malleable across records; record-labeled TDH2 restores confidentiality against parties who obtain other releases.
3. **An exact analysis** (Section 6): operating characteristics, showing that small audits cannot separate a committee one share short of t from a live one; sample sizes and sequential tests that certify a safety margin; a placement condition; a deadline budget; and, for custodians that fail and recover, the exact posterior risk given an audit history, a frequency floor and the longest audit interval meeting a risk target.
4. **An evaluation with labeled evidence** (Section 7): exact analysis, prespecified simulation, microbenchmarks and a 32-custodian testbed with a pre-registered replication, with negative results: a high benign alarm rate under the strictest rule, a selective-withholding counterexample, deadlines missed without network delay and slower batch verification.

None of the primitives is new, and a passed audit does not prove future availability.

Sections 2–7 give the model, properties, protocol, analysis and evaluation, and Sections 8–11 discuss them.

2 Preliminaries

Groups and sharing. Let $\mathbb{G}$ be a cyclic group of prime order ℓ with generator g in which the decisional Diffie–Hellman (DDH) problem is hard. The prototype uses ristretto255, a prime-order group built on Curve25519 [12], and keeps the quadratic residues modulo the RFC 7919 primes [13] (ffdhe2048, ffdhe3072; $\ell = (p-1)/2$, $g = 4$) as a baseline; we write groups multiplicatively, and q denotes only the acceptance threshold of an audit, and $\binom{a}{b} = 0$ when $b > a$ or $b < 0$. A secret

$x \in \mathbb{Z}_\ell$ is Shamir-shared with a polynomial f of degree $t-1$, $f(0) = x$; custodian $i \in \{1, \ldots, n\}$ holds $x_i = f(i)$ and publishes $Y_i = g^{x_i}$ [14], verifiable against Feldman commitments [5]. Any t distinct indices I have Lagrange coefficients λ_i^I with $\sum_{i \in I} \lambda_i^I x_i = x$.

Threshold Diffie–Hellman KEM. With epoch key $PK = g^x$, a sender draws $\rho \leftarrow \mathbb{Z}_\ell$, publishes $C_1 = g^\rho$ and derives a key-encryption key from PK^ρ and the record identifier. Any t partial decryptions $C_1^{x_i}$ determine $C_1^x = \prod_{i \in I} (C_1^{x_i})^{\lambda_i^I}$. This threshold ElGamal KEM [4] is semantically secure under DDH in the random-oracle model but malleable: anyone can turn C_1 into $C_1 g^\gamma$. Shoup and Gennaro's TDH2 binds a label and a proof of knowledge of ρ to the ciphertext, which makes it secure against chosen-ciphertext attacks [15]; KEYSTONE uses it for disputes (Section 5.2).

DLEQ proofs, hashing and beacon. A Chaum–Pedersen proof [16] shows $\log_g Y_i = \log_X D_i$: the prover picks $\nu \leftarrow \mathbb{Z}_\ell$, sends $a_1 = g^\nu$, $a_2 = X^\nu$, derives $c = H(g, Y_i, X, D_i, a_1, a_2, \mathsf{ctx})$ by the Fiat–Shamir transform [17] and returns $z = \nu + c\, x_i$; the verifier checks $g^z = a_1 Y_i^c$ and $X^z = a_2 D_i^c$. The request context ctx ties each proof to one request. H_G maps byte strings to $\mathbb{G} \setminus \{1\}$ with domain separation; we model H and H_G as random oracles. A public beacon outputs values that are unpredictable before publication [18].

3 System and threat model

An epoch E has n custodians placed in failure domains (providers, regions, software stacks), a threshold t, a key PK_E and registered public shares $Y_1, \ldots, Y_n$. A rollup worker encrypts each record r under a fresh data key K_r with AES-GCM, wraps K_r under the KEM, and publishes the result to a data-availability (DA) layer. A settlement layer, the *bulletin board*, records registrations, requests, deadlines and evidence and supplies an ordered public clock. A *combiner* inside the authorized destination collects partial decryptions over authenticated, confidential channels; it learns K_r for records it may open and is trusted not to disclose it. The board stores only commitments to responses with their validity status and timestamps, never enough to decrypt a record; canary responses may be public (Lemma 2). Figure 1 shows the three paths.

Assumption 1. Threshold confidentiality. During an epoch the adversary learns fewer than t current shares. DDH is hard in $\mathbb{G}$; H, H_G and the key-derivation function are independent, domain-separated random oracles; and AES-GCM is a secure authenticated-encryption scheme.

Assumption 2. Beacon. An audit's beacon value is unpredictable until its epoch, slot and committee are committed on the bulletin board, and the derived sample is uniform (without replacement, or within each domain quota for stratified samples) and independent of other audits' samples. The prototype derives the sample with a SHA-256 counter stream and rejection sampling, which we model as ideal.

Assumption 3. Bulletin board and data availability. The bulletin board is append-only and totally ordered and exposes each request to custodians within δ_b. The DA layer keeps committed ciphertexts and receipts retrievable, and record identifiers are unique.

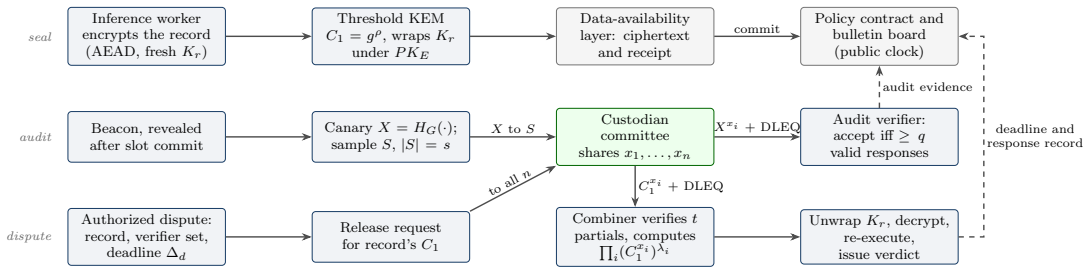


Figure 1. KEYSTONE architecture. Sealing, routine canary auditing and authorized dispute release are separate paths over one custodian committee. The audit uses a beacon-derived canary, never a production ciphertext; only an authorized dispute requests partial decryptions of a record's KEM component C_1. Dashed arrows are public evidence written to the bulletin board.

Assumption 4. Bounded delay, for deadline statements only. Messages between non-faulty parties arrive within δ over channels that authenticate the sending custodian. A non-faulty custodian produces a partial decryption with proof within c_p; the combiner verifies one proof within c_v and combines t partials within c_c.

Detection and confidentiality statements do not use Assumption 4. The adversary may crash custodians individually or by domain, cause share loss, send invalid or equivocal responses, issue unauthorized requests, act as a worker that commits records of its choice and obtains their releases, and control *selective* custodians that answer canaries but withhold targeted dispute responses. Corruption is static within an epoch. Out of scope are production key-generation security, combiners or verifiers that leak records they may open, arbitrary asynchronous partitions, and selective withholding beyond Observation 1. An audit shows that whoever answered knows x_i, not that the registered operator does, so share pooling and false domain declarations are invisible to it; and threshold custody replaces the one-honest-challenger assumption of optimistic rollups with an assumption on t of n custodians.

4 Dispute-key availability

Fix an epoch E, a record r with KEM component C_1, and a request time τ on the bulletin board.

Definition 1. Ciphertext availability. $\mathrm{CA}(r, \tau)$ holds if the committed encrypted record and its receipt are retrievable from the DA layer at τ.

Definition 2. Valid-share registration. $\mathrm{VSR}(E)$ holds if the registered public shares are consistent with one degree-$(t-1)$ polynomial whose constant term is the epoch secret, and every admitted custodian initially received the matching share.

Definition 3. Readiness. For a request req, an audit or a dispute on a named record, $\mathsf{Ready}_i(E, \tau, \mathsf{req}, \Delta) = 1$ if custodian i can access its current share, perform the requested

operation, produce a valid DLEQ proof and deliver the response by $\tau + \Delta$. Audit readiness and dispute readiness for each record are distinct predicates.

Definition 4. Audit-time readiness. For a sample S of s custodians and threshold $q \le s$, $\mathrm{AKR}(E, S, \tau)$ holds if at least q members of S deliver valid canary responses by the audit deadline.

Definition 5. Authorized decryptability and deadline liveness. After a valid authorization for r at τ, $\mathrm{AD}(r, E, \tau)$ holds if the combiner obtains valid partial decryptions of C_1 from at least t distinct custodians, and $\mathrm{DDL}(r, E, \tau, \Delta_d)$ holds if this happens by $\tau + \Delta_d$.

Definition 6. Pre-authorization confidentiality. PAC holds for r if no probabilistic polynomial-time adversary that satisfies Assumption 1, chooses two equal-length plaintexts for r and observes the sealed record and all audit transcripts, but no dispute releases, guesses which plaintext was sealed with non-negligible advantage. In the stronger PAC^+, the adversary may also commit records of its choice and obtain the release of every record other than r, where a *release* is the set of partial decryptions and the combined value delivered to the authorized destination, which the adversary may control for its own records.

Definition 7. Dispute-key availability. Fix a readiness model, a distribution over custodian failures, share loss and message delays over time. An epoch provides (ε, Δ_d)-DKA under it if for every committed record r and authorization time τ, $\Pr[\mathrm{DDL}(r, E, \tau, \Delta_d)] \ge 1 - \varepsilon$ over the readiness model and the protocol's randomness, and PAC^+ holds for every record before its authorization.

Definition 8. Catastrophic state and detection. An epoch is *catastrophic* at τ if fewer than t custodians are audit-ready. An audit *falsely accepts* if AKR holds in a catastrophic state; its *detection probability* is one minus its false-accept probability. Dispute readiness can be lower than audit readiness (Observation 1).

Remark 1. Timescale and epochs. Dispute windows last days and interactive moves hours, so DKA chiefly requires the key to stay serviceable throughout the window, while Δ_d matters for per-move timers. A record stays disputable after its committee rotates, so old committees must stay serviceable, or hand over shares by dynamic proactive sharing[19,20], and audits must cover every epoch with open records.

The next observation records what the weaker properties do not imply; each elementary witness blocks a tempting inference.

Observation 1. Non-implications. *Fix $1 \le t \le n$, $1 \le q \le s \le n$ and Δ_d.*
1. $\mathrm{CA}(r, \tau)$ *does not imply* $\mathrm{DDL}(r, E, \tau, \Delta_d)$.
2. $\mathrm{AKR}(E, S, \tau)$ *does not imply* $\mathrm{AD}(r, E, \tau')$ *for an authorized dispute at $\tau' \ge \tau$.*
3. For every m, m accepted audits at times $\tau_1 < \cdots < \tau_m$ do not imply $\mathrm{AD}(r, E, \tau')$ *for $\tau' > \tau_m$.*

Proof. (1) Let the DA layer retain the record while exactly $t - 1$ custodians are dispute-ready for r and the others have lost their shares. CA holds, but at most $t - 1$ valid partials exist, so AD and DDL fail. (2) Let all custodians be audit-ready and let $n - t + 1$ of them answer every

canary but withhold every response for r: AKR holds for every S, while at most $t-1$ custodians answer the dispute. (3) Take two executions with all custodians audit-ready up to τ_m; afterwards all stay ready in the first and $n-t+1$ lose their shares in the second. The audit histories are identical but AD holds at τ' only in the first, so without a model of how readiness evolves no predicate of the history implies AD.

Item (2) is why audit and dispute readiness are separate predicates; item (3) is why a repeated-audit guarantee needs a temporal model, static (Corollary 1) or stochastic (Section 6.2).

5 The KEYSTONE protocol

Setup and sealing. The committee is formed under identity, stake and failure-domain placement policy (Observation 2). A deployment would use an authenticated distributed key generation or publicly verifiable secret sharing[6,21,22]; the prototype uses a dealer with Feldman commitments as an explicit substitution boundary. To seal r under the label $L_r = (E, \mathsf{id}_r, \text{policy hash})$, the worker draws ρ, computes $C_1 = g^\rho$ and $\kappa = \mathrm{KDF}(PK_E^\rho, L_r)$, wraps K_r under κ, adds a TDH2 proof that binds L_r and the wrapped key to C_1 [15], encrypts the record under K_r with AES-GCM, and publishes the envelope with its DA pointer.

Readiness audit. The coordinator commits an epoch and audit slot. The next finalized beacon value β fixes a sample S of s custodians, uniform or stratified by domain, and everyone derives the canary $X = H_G(\texttt{KEYSTONE-CANARY} \parallel E \parallel \mathsf{slot} \parallel \beta)$. Each sampled custodian returns $D_i = X^{x_i}$ with a DLEQ proof that $\log_g Y_i = \log_X D_i$, bound to the request context and its index. The verifier checks membership, freshness, uniqueness per index, proof validity and the deadline, and accepts after q valid responses.

Authorized dispute. A dispute starts once the policy contract binds a record, epoch, authorized verifier and deadline Δ_d. Each custodian checks the authorization, the TDH2 proof and that the label is the authorized record's, and returns $C_1^{x_i}$ with a DLEQ proof bound to the dispute. The combiner verifies responses on arrival, off its network event loop, discards duplicates and invalid ones, stops at t valid partials, interpolates $C_1^x = PK_E^\rho$, re-derives κ, unwraps K_r and decrypts.

Transcript and evidence. For each request, the board records commitments to valid, invalid, equivocal, stale and unauthorized responses and the absence of missing ones, with timestamps. Invalid proofs and conflicting responses are self-contained cryptographic evidence; attributing a missing response to a custodian additionally requires Assumption 4. The prototype contract is a skeleton: an administrator supplies the sample and canary hash when opening an audit without an on-chain beacon check, records invalid-response evidence after off-chain verification, and can cancel requests. Theorem 1 therefore holds for the prototype only if the administrator samples as the protocol prescribes; deriving the sample on chain from the committed beacon and verifying proofs on chain are needed for a trustless deployment.

5.1 Security of the audit

Lemma 1. Correctness and knowledge soundness. *An honest response passes verification. In the random-oracle model a verifying response satisfies $D_i = X^{x_i}$ except with negligible probability,*

and the proof is a proof of knowledge: from any party that produces verifying responses for fresh canaries with non-negligible probability, an extractor obtains x_i. Any t verified partials of C_1 with distinct indices I combine to $\prod_{i\in I}(C_1^{x_i})^{\lambda_i^I} = PK_E^{\rho}$.

Proof sketch. Completeness and combination are direct computations. The protocol is special-sound, and with the Fiat–Shamir hash as a random oracle the forking lemma yields an extractor[16,17]. Full proofs of Lemmas 1–3 are in the supplementary material.

Knowledge soundness makes a passed audit meaningful: whoever passes canary audits with non-negligible probability holds x_i, and under Assumption 1 the adversary cannot obtain an honest custodian's share from public values without solving a discrete logarithm.

Lemma 2. Audit transcripts are simulatable. *Model H_G and H as random oracles. A simulator that knows only g and $Y_1, \ldots, Y_n$ answers all queries to H_G and H and produces the honest custodians' audit responses $(X, (D_i, \pi_i)_{i\in S})$ for any audit context and any S, including $S = \{1, \ldots, n\}$. With at most N simulated proofs and Q oracle queries, simulated and real views are within statistical distance NQ/ℓ, so observing audits changes any adversary's advantage against* PAC *by at most $2NQ/\ell$.*

Proof sketch. The simulator answers each fresh H_G query with g^a for a stored random a, so for an honest custodian and canary $X = g^a$ it can output the real response $D_i = Y_i^a = X^{x_i}$, and it produces each proof with the honest-verifier zero-knowledge simulator, programming H; a programmed input was queried before with probability at most Q/ℓ. Domain separation keeps this programming apart from a KEM reduction that programs the key-derivation oracle.

This is the standard simulation argument for threshold-coin and threshold-signature shares on hashed messages[10,11], which we restate for the audit setting. Lemma 2 holds even for adversarially chosen audit contexts and beacons. It is why the audit can exercise every sampled share without opening any record: combining t canary partials yields X^x for a hash-derived X, which equals a fixed record component only with negligible probability.

Lemma 3. No production substitution. *Model H_G as a random oracle. For a fixed C_1, a coordinator making Q queries to H_G finds an audit context whose canary equals C_1 with probability at most $Q/(\ell - 1)$, by a union bound over uniform oracle answers.*

Remark 2. The prototype hashes to ristretto255 with the standard map from a 64-byte hash[12]; the ffdhe baseline uses `expand_message_xmd`[23] before squaring. Binding request context into the Fiat–Shamir hash does not give the share isolation of context-dependent threshold decryption[24], and the lemmas do not cover side channels.

5.2 Confidentiality against releases of other records

Lemma 2 concerns audits only. An adversary that can also obtain releases, for example a worker that commits and disputes its own records, defeats the plain KEM.

Proposition 1. Malleability across records. *With the plain threshold Diffie–Hellman KEM,* PAC^+ *does not hold: an adversary that obtains the release of one record of its choice recovers K_r for any target record r.*

Proof. For the target's component C_1, the adversary commits a record r' with $C_1' = C_1 g^{\gamma}$ and obtains its release $C_1'^{x} = C_1^{x} PK_E^{\gamma}$. It computes $PK_E^{\rho} = C_1^{x} = C_1'^{x} PK_E^{-\gamma}$, derives the target's key-encryption key from PK_E^{ρ} and the target's public record identifier and metadata, and unwraps K_r. Already $\gamma = 0$, re-submitting C_1 under a new identifier, suffices.

The attack does not involve the audit. It is blocked by a threshold KEM whose custodians refuse components not bound to the authorized record, such as TDH2 with the record label[15] or context-dependent threshold decryption[24]; KEYSTONE uses TDH2.

Assumption 5. TDH2 key encapsulation. TDH2 used as a key-encapsulation mechanism, with $\kappa = \mathrm{KDF}(PK_E^{\rho}, L)$ in place of a one-time pad, is secure against chosen-ciphertext attacks in the threshold model of Shoup and Gennaro, who prove this for the encryption scheme under DDH in the random-oracle model[15].

Lemma 4. Record-bound releases. *Let the dispute KEM be TDH2 with label L_r, and let each custodian release $C_1^{x_i}$ only for a ciphertext whose TDH2 proof verifies and whose label is the authorized one. Under Assumptions 1 and 5 and static corruption,* PAC^{+} *holds for every record, and Lemma 2 continues to hold.*

Proof sketch. A release of a record $r' \neq r$ answers a decryption-share query for a valid ciphertext with label $L_{r'} \neq L_r$. Because the label enters the TDH2 proof, a re-labelled or mauled C_1 either fails verification and is refused, or is a valid ciphertext other than the challenge, which the chosen-ciphertext game of Assumption 5 answers. The target's key-encryption key is therefore pseudorandom, and AES-GCM hides the wrapped key and then the record. The canary path is unchanged and its oracles are domain separated, so the simulation of Lemma 2 composes with the TDH2 reduction.

The cost is one ciphertext-proof check per custodian per dispute (0.116 ms in ristretto255, Table 6). The prototype's custodians enforce both checks, and a test reproduces the attack of Proposition 1 once they are removed.

6 Analysis

Theorem 1. False-accept probability. *Suppose exactly k custodians are audit-ready throughout one audit, the sample S of size s is uniform without replacement and independent of readiness (Assumption 2), and acceptance requires q valid responses. Then, up to the negligible soundness error of Lemma 1,*

$$\Pr[\mathsf{accept} \mid k] = \sum_{j=q}^{\min(s,k)} \frac{\binom{k}{j}\binom{n-k}{s-j}}{\binom{n}{s}}. \tag{1}$$

This is non-decreasing in k, so over catastrophic states ($k \leq t-1$) it is maximized at $k = t-1$, and for $q = s$

$$\varepsilon_a = \max_{k<t} \Pr[\mathsf{accept} \mid k] = \frac{\binom{t-1}{s}}{\binom{n}{s}}. \tag{2}$$

Proof. The number of ready custodians in S is hypergeometric (n, k, s), and by Lemma 1 the audit accepts exactly when it is at least q, up to negligible error. If $R \subset R'$, then $|S \cap R| \leq |S \cap R'|$ for every S, which gives monotonicity; for $q = s$ only $j = s$ contributes.

For $n = 32$, $t = 22$, $s = q = 8$, $\varepsilon_a = \binom{21}{8}/\binom{32}{8} = 203{,}490/10{,}518{,}300 = 0.0193$, so a single audit detects the worst catastrophic state with probability 0.9807 under a static, non-selective fault model. Section 6.1 shows why this number alone is misleading. Supplementary Figure S1 shows detection against s.

Corollary 1. Repeated audits under a static state. *If the ready set is fixed and catastrophic for m audits whose samples are independent (Assumption 2), all m falsely accept with probability at most ε_a^m.*

Proof. Given the fixed ready set, the acceptance events are independent, each with probability at most ε_a by Theorem 1.

By Observation 1(3), the static hypothesis cannot be dropped without a temporal model.

For samples stratified by domain, the per-domain counts of ready sampled custodians are independent hypergeometric variables, so acceptance is the probability that their sum reaches q (supplementary Proposition S1); the worst catastrophic state then depends on how the $t - 1$ ready custodians are spread across domains.

Observation 2. Placement cap. *With no failures other than whole-domain outages, no single domain outage prevents reconstruction if and only if every domain has at most $n - t$ custodians, since an outage of a domain of size d leaves $n - d$.*

The baseline places four domains of eight under the cap $n - t = 10$: one outage is survivable, two leave $16 < 22$ custodians.

Observation 3. Benign pass probability. *If each custodian is independently unavailable with probability f and the sample is independent of availability, the audit passes with probability* $\Pr[\mathrm{Bin}(s, 1 - f) \geq q]$*, since the sampled custodians' availability indicators are independent.*

The strict rule $q = s$ that maximizes detection also turns every benign miss into an alarm. For $s = q = 8$ and $f = 0.10$ a healthy committee passes only with probability $0.9^8 = 0.430$, although t of 32 shares are reachable with probability $\Pr[\mathrm{Bin}(32, 0.9) \geq 22] = 0.99983$. Table 1 and supplementary Figure S2 give the trade-off. At $s = 8$, $q = 7$ lowers detection to 85.9% and raises the benign pass rate at $f = 0.10$ only to 81.3%; a larger sample helps: $s = 16$, $q = 14$ detects 98.8% with benign pass rates of 95.7% at $f = 0.05$ and 78.9% at $f = 0.10$. This post hoc analysis is design guidance, not a prespecified result.

6.1 Operating characteristic and sequential audits

Worst-case detection hides how weak a small audit is near the threshold. Table 2 gives the exact operating characteristic: with $s = q = 8$ the audit accepts a committee one share short of t with probability 1.9%, a live committee with no spare share with 3.0%, and one with four spare shares with only 14.9%. Only the full probe ($s = n$) separates $k = t - 1$ from $k = t$ at 5% error rates, so

Table 1. Exact acceptance-rule trade-off at $n = 32$, $t = 22$: worst-case detection ($k = t - 1$, Theorem 1) versus the probability that a healthy committee passes under IID unavailability f (Observation 3). Post hoc analysis.

		Worst-case	Benign audit pass under IID unavailability		
s	q	detection	$f = 0.05$	$f = 0.10$	$f = 0.20$
8	6	57.5%	99.4%	96.2%	79.7%
8	7	85.9%	94.3%	81.3%	50.3%
8	8	98.1%	66.3%	43.0%	16.8%
12	10	89.6%	98.0%	88.9%	55.8%
12	11	98.2%	88.2%	65.9%	27.5%
12	12	99.9%	54.0%	28.2%	6.9%
16	13	93.2%	99.3%	93.2%	59.8%
16	14	98.8%	95.7%	78.9%	35.2%
16	15	99.9%	81.1%	51.5%	14.1%

Table 2. Exact operating characteristic: probability that an audit with sample size s and threshold q accepts when k of $n = 32$ custodians are ready ($t = 22$, uniform sampling, Theorem 1).

		Pr[audit accepts] with k ready custodians ($n = 32$, $t = 22$)					
s	q	$k = 21$	$k = 22$	$k = 24$	$k = 26$	$k = 28$	$k = 30$
8	8	1.9%	3.0%	7.0%	14.9%	29.5%	55.6%
8	7	14.1%	19.3%	33.3%	52.4%	74.6%	94.4%
16	14	1.2%	2.7%	11.0%	32.7%	70.0%	100.0%
16	13	6.8%	12.6%	34.3%	67.3%	94.9%	100.0%
24	20	0.1%	0.5%	8.2%	47.6%	100.0%	100.0%

sampled guarantees must be stated for a safety margin m, testing $k \le t - 1$ against $k \ge t + m$. With error rates of 5% each, a single audit needs $s = 25$ ($q = 19$) for $m = 4$, $s = 20$ ($q = 16$) for $m = 6$ and $s = 14$ ($q = 12$) for $m = 8$. A sequential probability ratio test across audits[25], valid under the static state of Corollary 1, is far cheaper: for $m = 6$ it decides on average after 2.3 audits for a failed committee and 1.1 for a live one with $s = 16$, $q = 14$, and after 9.8 and 4.6 with $s = q = 8$ (Wald's approximations). Audits should therefore be specified by the margin they certify and their error rates, not by worst-case detection.

Proposition 2. Deadline budget. *Under Assumptions 3 and 4, suppose at least t custodians are non-faulty, hold their shares and follow the protocol, and the combiner is work-conserving, verifies serially and verifies each authenticated index at most once. Then a dispute requested at τ obtains t valid partials by $\tau + \delta_b + c_p + \delta + n\,c_v$, and the key is recovered c_c later. Hence $\Delta_d \ge \delta_b + c_p + \delta + n\,c_v$ suffices for* DDL*, and $n\,c_v$ may be replaced by $t\,c_v$ if no invalid response precedes the t-th valid one.*

Proof. Each of the t correct custodians' responses arrives by $T' = \tau + \delta_b + c_p + \delta$. Authenticated channels stop an adversary from occupying a correct custodian's index, so the work-conserving combiner verifies at most n responses and finishes by $T' + n\,c_v$, or by $T' + t\,c_v$ if no invalid response comes first.

On a production settlement layer, δ_b is a block or finality time and can dominate. With random delays, a per-message quantile for δ does not budget for the slowest of the t responses needed:

Corollary 2. Stochastic deadline. *Suppose n_ℓ custodians can answer, each response is delivered independently with probability p_ℓ after a round trip with distribution function F, and the combiner finishes within B after the t-th valid response. Then* $\Pr[\mathrm{DDL}(r, E, \tau, \Delta_d)] \geq \Pr[\mathrm{Bin}(n_\ell, p_\ell F(\Delta_d - B)) \geq t]$.

Proof sketch. The number of responses delivered within $\Delta_d - B$ is $\mathrm{Bin}(n_\ell, p_\ell F(\Delta_d - B))$; if it reaches t, the dispute completes by Δ_d.

6.2 Readiness over time

We now let custodians fail and recover. Each custodian is an independent two-state Markov chain with failure rate ϕ and repair rate ψ, so it is up a fraction $v = \psi/(\phi + \psi)$ of the time; in a variant, each of four domains of eight also fails as a unit at rate ϕ_d and recovers at rate ψ_d. Audits run every T with a fresh uniform sample and only observe: they trigger no repair. Faults are non-selective, so a custodian is audit-ready exactly when it is dispute-ready, and disputes arrive independently of the process. Let $Z(\tau)$ be the number of ready custodians, $A(k)$ the acceptance probability of Eq. (1), ω the stationary law of Z, and $\mathbf{1}_{<t}$ the indicator column of $Z < t$; vectors are rows over $k \in \{0, \ldots, n\}$.

Theorem 2. Risk given an audit history. *Z is a Markov chain with $P_u(k, k') = \sum_i \mathrm{b}(i; k, p_\uparrow(u))\, \mathrm{b}(k' - i; n - k, p_\downarrow(u))$, where* b *is the binomial mass function, $p_\uparrow(u) = v + (1 - v)e^{-(\phi+\psi)u}$ and $p_\downarrow(u) = v(1 - e^{-(\phi+\psi)u})$. For audit outcomes $o_1, \ldots, o_j$, let $\alpha_1 \propto \omega \odot e_{o_1}$ and $\alpha_i \propto (\alpha_{i-1} P_T) \odot e_{o_i}$, normalized, with $e_{\mathsf{acc}} = A$ and $e_{\mathsf{rej}} = 1 - A$. A dispute at $jT + u$, $0 \leq u < T$, fails to reach t with probability $\alpha_j P_u \mathbf{1}_{<t}$ given $o_1, \ldots, o_j$. With domain shocks, the same holds for the chain of per-domain pairs (domain up, members up).*

Proof sketch. From any configuration with k custodians up, after time u each up custodian is up with probability $p_\uparrow(u)$ and each down one with $p_\downarrow(u)$, independently, so Z is strongly lumpable with kernel P_u. A fresh uniform sample makes each outcome conditionally independent of everything else given Z at its audit, with emission A, so the history is a hidden Markov model and the recursion is the forward algorithm.

After an unbroken run of accepted audits, α_j converges from any start to the normalized Perron vector α^* of $P_T \operatorname{diag}(A)$ on the states the audit can accept, where it is entrywise positive for $T > 0$. We call $R(T) = \sup_{0 \leq u < T} \alpha^* P_u \mathbf{1}_{<t}$ the *posterior risk*: the failure probability a dispute faces, at the worst point of an audit cycle, after a long clean audit record.

Theorem 3. Undetected failure. *Let $U(T) = \frac{1}{T} \int_0^T (\omega \odot A) P_u \mathbf{1}_{<t}\, du$, the long-run probability that a dispute fails and the most recent audit accepted. With or without domain shocks, U is non-decreasing in T and $U(T) \geq U_0 = \sum_{k<t} \omega(k) A(k)$, its limit as $T \to 0$. Without domain shocks,*

$$U(T) \leq A(t-1)\, \omega(Z < t) + \sum_{k \geq t} \omega(k) A(k) \Pr[\mathrm{Bin}(k, p_\uparrow(T)) < t].$$

Proof sketch. The chain is reversible, moves one custodian or domain at a time, and is stochastically monotone because $p_\uparrow \geq p_\downarrow$. Let $g(u) = (\omega \odot A)P_u\mathbf{1}_{<t}$. Then $g'(u)$ is minus the Dirichlet form of A and $P_u\mathbf{1}_{<t}$; A is increasing and P_u keeps $\mathbf{1}_{<t}$ decreasing, so every term has the sign that makes $g' \geq 0$. Hence U, the mean of g over $[0, T]$, is non-decreasing and at least $g(0) = U_0$. For the bound, use $A(k) \leq A(t-1)$ for $k < t$, and for $k \geq t$ note that the custodians up at the audit and still up at u number $\mathrm{Bin}(k, p_\uparrow(u))$, lower-bound $Z(u)$, and decrease in u.

Since audits trigger no repair, the failure probability itself is $\omega(Z < t)$; U measures how much of it the latest audit fails to flag. A target $\varepsilon < U_0$ cannot be met by auditing more often, only by a stronger design or a larger availability margin. Otherwise the longest admissible interval $T^*(\varepsilon) = \sup\{T : U(T) \leq \varepsilon\}$ is the cheapest schedule, found by bisection because U is monotone, and the bound gives a conservative interval in closed form. Full proofs are in the supplementary material.

7 Evaluation

We compare the audit with simpler checks (Q1), separate audit outcomes from dispute success under failures (Q2), test deadline liveness (Q3), measure cost (Q4) and evaluate readiness over time (Q5).

Method. Table 3 lists every experiment, and each result carries one of four evidence labels. *Exact* values are closed forms or recursions under a declared model. *Simulation* results come from cells whose parameters, endpoints and seeds were fixed and hashed before execution; the three Q2 cells have 131,072 draws each, above the 73,778 that Hoeffding's inequality requires for a proportion's half-width of 0.005 at 95% confidence [26], and where an exact value exists simulation serves as an implementation check. *Microbenchmarks* are single-process medians, and *emulation* runs 32 custodian processes on one host. A first prototype (R1: ffdhe2048, serial verification on the network loop) produced Q1, Q2 and the comparison column of Q3; the second (R2: ristretto255, TDH2, verification off the loop) followed a second design hashed before any R2 run. Proportions carry 95% Wilson intervals [27].

7.1 Readiness baselines (Q1)

A heartbeat is answered even by a custodian that lost its share, registration is checked once, and decrypting a real record discloses it (supplementary Table S4 compares these checks). A full-committee canary probe ($s = n$) detects every static catastrophic state at four times KEYSTONE's cost and is affordable once per epoch at $n = 32$. Sampling pays off when audits are frequent, committees are large or responses are recorded on chain: at 57,882 gas per response (supplementary Table S2), responses cost about 0.46 million gas per audit for $s = 8$ and 1.85 million for $s = 32$.

7.2 Robustness under failures (Q2)

In the simulation cells (Table 4), 10% independent unavailability leaves the committee reconstructable in 99.98% of draws while the strict audit passes in only 43.1%: a sensitive alarm, not a health score. Under 20% domain outages, stratified sampling raised conditional detection

Table 3. Experimental configuration. The baseline committee is $n = 32$, $t = 22$, $s = q = 8$ with four failure domains of eight custodians.

Experiment family	Parameters	Execution and evidence label
Static bound (Eq. 2)	$n \in \{16, 32, 64, 128\}$, $t = n - \lfloor (n-1)/3 \rfloor$, $s \in \{1, \ldots, 16\}$, $q = s$; baseline $n = 32, t = 22, s = q = 8$	Exact closed form
Simulation cells	$n = 32, t = 22, s = q = 8$, 4 domains of 8. IID: $p_{\text{off}} = 0.10$. STRAT: $p_{\text{dom}} = 0.20$, $p_{\text{off}} = 0.02$, uniform vs. 2-per-domain quota. SW: $w = 11$	32 seed blocks × 4,096 draws = 131,072 per cell; seeds frozen before execution; simulation
Correlation model	$f \in \{0, 0.025, \ldots, 0.40\}$; IID vs. 2/4/8 equal domains	Exact; declared synthetic model
Acceptance rule	$s \in \{8, 12, 16\}$, $q \in [\lceil s/2 \rceil, s]$, IID $f \in \{0.05, 0.10, 0.20\}$	Exact; post hoc
Cryptographic cost	R1: ffdhe groups, $n \leq 128$. R2: ristretto255 and ffdhe, 1,000 runs	Microbenchmark, CPython 3.13, Apple M4 Pro
Deadline testbed	R2 and R2b (fresh paired seeds): ristretto255, TDH2, $n = 32, t = 22$; 8 profiles × 300 traces × 2 variants; lognormal delay (median 0/40/120 ms), loss, crashes, domain outage	32 processes on one host; emulation
Temporal readiness	MTBF 7/30/90 d, MTTR 1/6/24 h, T 1/6/24 h, (s, q) $(8, 8)/(16, 14)/(24, 20)$; domain shocks	Exact, synthetic; 3-cell simulation check

Table 4. Simulation cells, 131,072 draws each, 95% Wilson intervals. Reconstructable means at least t custodians can serve the dispute.

Scenario	Audit pass	Reconstructable
IID 10% unavailable	43.1% [42.8, 43.3]	99.98% [99.97, 99.99]
Domain outage 20%, uniform sample	37.3% [37.0, 37.6]	81.3% [81.1, 81.6]
Domain outage 20%, stratified sample	34.9% [34.7, 35.2]	81.3% [81.1, 81.6]
11 selective withholders	100.0% [99.997, 100.000]	0.0% [0.000, 0.003]

of catastrophic draws from 99.89% to 100.00% and lowered the benign pass rate from 37.3% to 34.9%. The prespecified endpoint, the paired difference in detected catastrophes per draw, was 0.00021 with Hoeffding half-width 0.0075, too conservative to resolve an effect this small; a post hoc exact sign test on the 28 discordant pairs, all favoring stratification, gives $p = 7.5 \times 10^{-9}$. With eleven selective withholders, audits passed in 100.0% of draws and disputes succeeded in 0.0%, reproducing the witness of Observation 1(2).

Placement matters more (Figure 2). At per-unit unavailability $f = 0.10$, an IID committee reconstructs with probability 99.98%, four domains of eight (under the cap of Observation 2)

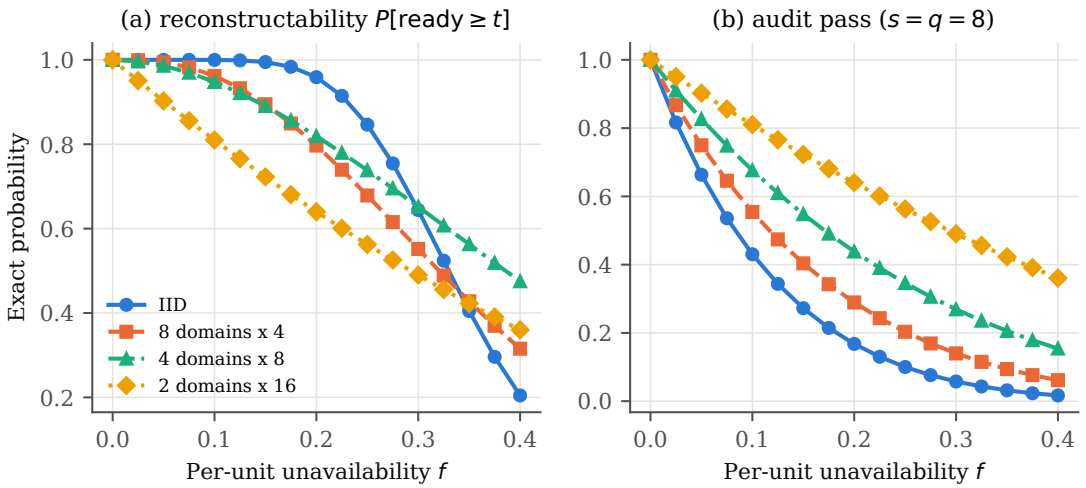


Figure 2. IID versus domain-correlated unavailability at $n = 32$, $t = 22$. A domain fails with probability f and takes all its members; in the IID model each custodian fails independently with probability f. (a) Reconstructability. (b) Audit pass with $s = q = 8$. Exact; a 20,000-draw Monte Carlo check agrees within 0.009.

with 94.8%, and two domains of sixteen, which violate it, with only 81.0%. Field studies show that correlated failures dominate real unavailability [28].

7.3 Deadline liveness in an emulated testbed (Q3)

Thirty-two custodian processes, each a TCP server holding only its own ristretto255 share, and one combiner run on one host. Each trace runs one canary audit ($s = q = 8$) and one dispute over a fresh 1 KiB record; the combiner verifies DLEQ proofs in a worker thread and stops at $t = 22$ valid partials. Delays are lognormal on both legs, lost messages are not retransmitted, and crashed custodians never answer. Two runs of eight profiles × 300 traces × two verification variants used different deadlines (Table 5). R2 used $\Delta_d^* = c_p + 2\delta_{99} + t\,c_v + c_c$, Proposition 2 with each leg's 99th-percentile delay and the medians of Table 6, fixed before its run. R2b, designed after R2 and fixed before any of its traces, used fresh seeds shared by both variants and the deadline Δ_{os} at which Corollary 2 gives 0.99, with B calibrated below.

R2. Five of the seven reconstructable profiles succeeded in every trace, and eleven crashes ($21 < t$) defeated every dispute. With delay, completion followed the network, with medians from 106 to 601 ms, against 3.69–3.88 s in R1, where serial verification on the network loop took about 3 s. Two profiles missed. Without delay, no dispute met the 4.36 ms deadline (median 6.7 ms): the formula counts cryptographic work only, not message handling among 33 processes. With ten crashes all 22 survivors must answer, so the slowest of 22 round trips decides: 97.3%

Table 5. Emulated testbed: 32 custodian processes on one host, ristretto255 and TDH2, 300 traces per profile, variant and run. Success requires the correct plaintext by the run's deadline: Δ_d^* for R2, Δ_{os} (Corollary 2, calibrated B) for R2b, each fixed before its run; R2b serial with 95% Wilson interval; medians over R2b serial disputes that succeeded; R1 is the first prototype (ffdhe2048).

	R2 (frozen deadline)			R2b (corrected deadline, fresh traces)				R1
Profile	Δ_d^* (ms)	serial	batch	Δ_{os} (ms)	serial	batch	median (ms)	median (ms)
control	4.4	0.0%	0.0%	7.1	83.3% [78.7, 87.1]	99.3%	6.9	3818
regional	260	100.0%	100.0%	128	98.7% [96.6, 99.5]	87.7%	108	3841
global	974	100.0%	100.0%	408	98.7% [96.6, 99.5]	98.3%	325	3875
global loss2	974	100.0%	100.0%	435	98.7% [96.6, 99.5]	98.0%	340	3829
global crash10	974	97.3%	94.7%	1176	99.0% [97.1, 99.7]	99.0%	602	3692
global crash11	974	0.0%	0.0%	–	0.0% [0.0, 1.3]	0.0%	–	–
global domain outage	974	100.0%	100.0%	663	99.3% [97.6, 99.8]	99.0%	464	3700
heavy tail	3918	100.0%	100.0%	785	100.0% [98.7, 100.0]	100.0%	464	3771

and 94.7% succeeded, and Corollary 2 with the frozen parameters, applied post hoc, predicts 96.3%.

Calibration and R2b. On the same harness without cryptography, 22 responses took 1.23 ms at the median, 37 μs of it per hand-off to the verification thread; the full dispute path without delay took 6.83 ms, with 99th percentile $B = 7.09$ ms. In R2b, serial verification met the criterion, a Wilson upper bound of at least 0.99, in 7 of eight profiles, although Δ_{os} is far tighter than Δ_d^* (408 against 974 ms under global delay), and at the R2 deadline the ten-crash profile again succeeded in 96.3%. It missed without delay: 83.3% [78.7, 87.1], as the median rose from 6.83 to 6.9 ms between runs, so a deadline without network slack is fragile to small host drift.

Batch verification. In the paired R2b design, batch verification was 8–12 ms slower at the median in every delayed profile (sign tests, $p < 10^{-74}$), plausibly because it starts only after the last needed arrival while serial verification overlaps with arrivals, and 0.9 ms faster without delay, where responses arrive together and it saves hand-offs. It therefore met the control deadline (99.3%) but missed the regional one (87.7%), contrary to the expected absence of a difference. Audit pass rates fell inside the Wilson interval of their exact expectation in 16 and 16 of 16 profile and variant pairs.

7.4 Cost and scalability (Q4)

Table 6 gives per-operation medians of the R2 prototype. In ristretto255 a partial decryption with proof takes 0.086 ms and its verification 0.15 ms, so the term $t\,c_v$ of Proposition 2 is 3.4 ms, against 1,062 ms for R2's own ffdhe2048 code and about 2,980 ms for R1's; combining 22 partials, unwrapping and decrypting takes 0.92 ms. Batch verification of 22 proofs by a random linear combination runs at 0.93 times the speed of serial verification in ristretto255, where scalar multiplications dominate either way, and at 1.83 and 2.37 times in ffdhe2048 and ffdhe3072. Supplementary Table S1 gives R1 costs for committees of up to 128.

Table 6. Median wall time per operation in the R2 prototype, 1,000 timed runs, CPython 3.13 on an Apple M4 Pro; ristretto255 through libsodium, RFC 7919 groups[13] in pure Python. Batch verification uses 128-bit random coefficients.

Operation (median, ms)	ristretto255	ffdhe2048	ffdhe3072
Canary derivation H_G	0.011	0.020	0.034
Partial decryption + DLEQ proof	0.086	49.53	160.8
DLEQ verification	0.152	48.26	140.0
Batch verification of $t = 22$ proofs	3.61	580.7	1301.6
Seal 1 KiB, plain KEM	0.041	32.38	103.9
Seal 1 KiB, TDH2	0.106	83.29	266.5
TDH2 ciphertext-proof check	0.116	47.85	137.6
Combine t partials, unwrap, decrypt	0.923	244.9	729.6

Table 7. Readiness over time, exact under a declared synthetic model, $n = 32$, $t = 22$. Scenarios give custodian MTBF / MTTR and, for shocks, the domain MTBF (domain MTTR 6 h). U: probability that a dispute fails and the last audit accepted; U_0: its floor; T^*: longest interval with $U \le 10^{-4}$, by bisection over $[0.25\,\text{h}, 30\,\text{d}]$ (none: infeasible at 0.25 h).

		U, $T = 6$ h		U_0		T^* (h), $\varepsilon = 10^{-4}$	
Scenario	$\Pr[Z < t]$	(8,8)	(24,20)	(8,8)	(24,20)	(8,8)	(24,20)
IID, 7 d / 24 h	1.2×10^{-3}	3.7×10^{-5}	2.4×10^{-5}	2.1×10^{-5}	6.2×10^{-7}	22.8	13.3
IID, 30 d / 24 h	2.7×10^{-9}	1.0×10^{-10}	8.1×10^{-11}	5.1×10^{-11}	1.7×10^{-12}	≥720	≥720
Shock 365 d, 30 d / 6 h	5.5×10^{-6}	1.2×10^{-6}	1.6×10^{-6}	5.5×10^{-8}	1.8×10^{-9}	≥720	≥720
Shock 90 d, 30 d / 6 h	5.7×10^{-5}	1.1×10^{-5}	1.3×10^{-5}	2.6×10^{-7}	7.1×10^{-9}	≥720	≥720
Shock 90 d, 7 d / 24 h	7.7×10^{-3}	6.9×10^{-4}	1.8×10^{-3}	1.1×10^{-4}	2.5×10^{-6}	none	0.3

7.5 Readiness over time (Q5)

We evaluated Theorems 2 and 3 on a synthetic grid of 243 cells (Table 3); Table 7 shows five scenarios. The bound of Theorem 3 held in all 81 shock-free cells. An event simulation without closed forms matched 29 of the 30 quantities with observed events within 99% Wilson intervals (3 more, near 10^{-16}, saw no events). Three findings stand out. First, correlated failures dominate: with custodian MTBF 30 d and MTTR 6 h, $\Pr[Z < t]$ is 1.4×10^{-15} for independent custodians and 5.7×10^{-5} when each domain fails every 90 days on average. Second, the floor binds: with domain shocks, custodian MTBF 7 d and MTTR 24 h, $U_0 = 1.1\times10^{-4}$ for $(8, 8)$, so no audit interval reaches $\varepsilon = 10^{-4}$, and $(24, 20)$ reaches it only with audits every 15 minutes, at the edge of the searched range. Third, larger samples are not uniformly better: $(24, 20)$ lowers the floor by orders of magnitude, yet its U exceeds that of $(8, 8)$ in all 54 shock cells and 18 of 27 shock-free ones, each time because it accepts more often; U ignores the benign alarms the strict rule raises instead. For independent custodians with MTBF 7 d, MTTR 24 h and $(8, 8)$, the longest interval for $\varepsilon = 10^{-4}$ is 22.8 h under U, 12.2 h under the posterior risk R, and a conservative 3.1 h from the bound.

8 Discussion

Five design consequences follow. First, specify audits by the safety margin they certify, with error rates at $k \leq t-1$ and $k \geq t+m$ (Section 6.1). Second, placement under the cap $n-t$ matters more than the sampling policy, and correlated domain failures dominate the risk over time (Q5). Third, schedule audits from a risk target: Theorem 3 gives the longest admissible interval, and below the floor U_0 only a stronger design helps. Fourth, deadlines must budget for the ledger delay, host overhead and the slowest of the t needed responses (Corollary 2), not a per-message quantile; with elliptic-curve verification off the network loop, the network governs completion, and batch verification helps only when responses arrive together (Q3). Fifth, if untrusted parties can obtain releases, the dispute path must bind each release to its record (Proposition 1, Lemma 4).

Invalid proofs and equivocation are transferable evidence, whereas attributing non-response depends on timing[29], so slashing should rest on transferable evidence. The audit cannot detect selective withholding, because canaries are unrelated to records by design (Lemma 2); making them indistinguishable from record components would stop observers from checking that an audit opened no record (Lemma 3), so resistance needs another mechanism, such as unpredictable dispute routing or economic commitments enforced by the deadline transcript.

9 Related work

Optimistic and verifiable computation. Optimistic rollups rely on an honest challenger to force correct results on chain[1]; opML applies this to machine-learning inference[2], and EigenAI re-executes private inference in a trusted environment after a threshold key release[3]. These systems assume the challenger can read the disputed inputs; with encrypted inputs this becomes the requirement that keys be serviceable before the deadline, which KEYSTONE audits without changing the dispute game.

Threshold decryption and key management. Threshold cryptosystems[4], their chosen-ciphertext-secure variants[15] and DLEQ proofs[16,17] provide the release mechanism. Verifiable and publicly verifiable secret sharing and distributed key generation[5,6,21,22] make shares consistent when created, and proactive and dynamic sharing refreshes them[7,19,20]; a refresh also exercises every share, but it is infrequent and costly. A canary response is structurally a threshold-coin share[10] or a threshold-signature share on a hashed message[11]; our contribution is its use as a sampled readiness test and what that test can and cannot certify. Context-dependent threshold decryption isolates shares across contexts[24].

Threshold release on ledgers. CALYPSO releases on-ledger secrets to authorized readers through a threshold committee with an auditable access log[30], and ETHTID provides threshold disclosure on Ethereum[31]. Encrypted mempools face the same concerns of committees, epochs, collusion and timely decryption[32–34]. FairBlock has a committee exchange a fixed number of identity-based decryption shares per block[35], and F3B, building on CALYPSO, has a committee release a per-transaction key once the transaction is final[36]. In both, timely decryption depends on a threshold of members serving the release, the property DKA makes explicit; the descriptions we found specify the release path but no pre-release readiness audit. UQ-Vote evaluates

liveness of consensus-authorized threshold decryption under faults[37]. These ideas are established; KEYSTONE adds the deadline-bound property that release must satisfy for disputes, an audit that measures it without opening records, and an analysis of what passing it implies.

Proofs of storage, custody and heartbeats. Proofs of retrievability and provable data possession spot-check stored files and bound the probability that a server missing data passes[8,38], also under a mobile adversary[39]. KEYSTONE's audit belongs to this family, with a proof of knowledge (Lemma 1) and sampling bounds (Theorem 1). It differs in that possession is not service (Observation 1), in that confidentiality forces canary responses to be useless for decrypting records (Lemmas 2 and 3), and in that DKA is a threshold, target-specific and deadline-bound property. Proof-of-custody proposals penalize validators that do not hold attested data[40]. The closest deployed analogue is the tBTC v2 heartbeat, in which each threshold wallet's signing group signs periodic messages and funds leave a wallet that fails one[41]; KEYSTONE instead samples custodians with exact operating characteristics, audits decryption shares under a confidentiality constraint, and models readiness over time. Data-availability proofs and sampling[9,42] concern public data, which KEYSTONE takes as given. Supplementary Table S3 compares these mechanisms.

Accountability. Slashing conditions[43] and BFT forensics[44] attribute safety violations with transferable evidence, threshold traitor tracing addresses coalitions that leak decryption capability[45], and attributing liveness failures depends on timing[29].

10 Threats to validity and limitations

Novelty. The components are not new; our claim covers the property decomposition, the confidentiality-preserving audit and its analysis, and the labeled evaluation, and it rests on a bounded literature search.

Internal validity. Both designs were hashed before execution, but the hashes are self-recorded and no independent party reviewed the designs; the acceptance-rule and operating-characteristic analyses were added afterwards. R2 and R2b logged thirteen deviations, among them a deadline correction before any testbed run and a restart after the host ran out of ephemeral ports; none changed an endpoint or an analysis in response to outcomes, and the corrected deadline was tested only on fresh R2b traces.

Construct and external validity. The audit measures present sampled serviceability, not future cooperation. The temporal model assumes independent exponential failures and repairs with synthetic rates, failure models are synthetic, the testbed uses loopback TCP with synthetic delays, and geo-distributed effects, ledger delay and clock skew are not measured.

Prototype scope. A dealer replaces distributed key generation; the contract trusts an administrator for sampling, invalid-response evidence and cancellation; neither code nor contract has been externally audited; the lemmas hold in the random-oracle model with static corruption; and Lemma 4 and the temporal results are given with proof sketches here and full proofs in the supplementary material.

11 Conclusion and future directions

Ciphertext availability does not ensure that the threshold service needed for an encrypted-rollup dispute answers in time. We defined dispute-key availability and the weaker properties it is confused with, showed that a beacon-derived canary audit exercises real shares while its transcripts are simulatable from public data, gave its exact operating characteristics, sample sizes and sequential tests that certify a safety margin, a placement condition and a deadline budget, and showed that confidentiality against parties who obtain other releases requires binding releases to records, as record-labeled TDH2 does. For custodians that fail and recover, an exact recursion gives the risk that a dispute fails after any audit history, and a floor shows when auditing more often cannot help. The evaluation confirmed each separation: a small strict audit barely distinguishes a failed committee from a live one and alarms often on healthy committees, selective withholders pass every audit and defeat every dispute, correlated domain failures dominate risk over time, and with elliptic-curve verification off the network loop the network governs dispute completion; an order-statistic deadline held in a pre-registered replication except without network delay, where host drift of a few percent defeats any deadline without slack. Future work should address selective withholding, which the audit provably cannot detect alone; readiness models calibrated on operational data; and a production instantiation with distributed key generation and realistic ledger delays, evaluated across regions.

Statements and Declarations

Ethical considerations. This computational study uses synthetic data, simulation, prototype outputs and smart-contract tests, and involves no human participants, personal data or live systems; ethical approval was not required.

Consent to participate. Not applicable.

Consent for publication. Not applicable.

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Data availability. Code, frozen designs with SHA-256 hashes, deviation logs, raw outputs and the scripts that generate every table, figure and in-text number are provided as anonymized supplementary material. A public repository link is on the title page.

Declaration of AI use. AI assistance was used for drafting text, integrating code, scripting experiments and generating figures. The author(s) verified the scientific content, the proofs and every reported number against the generated outputs, and take full responsibility for the article.

References

1. Harry Kalodner, Steven Goldfeder, Xiaoqi Chen, S. Matthew Weinberg, and Edward W. Felten. Arbitrum: Scalable, private smart contracts. In *27th USENIX Security Symposium (USENIX*

Security 18), pages 1353–1370. USENIX Association, 2018. ISBN 978-1-939133-04-5. URL https://www.usenix.org/conference/usenixsecurity18/presentation/kalodner.

2. KD Conway, Cathie So, Xiaohang Yu, and Kartin Wong. opML: Optimistic machine learning on blockchain. arXiv preprint arXiv:2401.17555, 2024. URL https://arxiv.org/abs/2401.17555.
3. David Ribeiro Alves, Vishnu Patankar, Matheus Pereira, Jamie Stephens, Nima Vaziri, and Sreeram Kannan. EigenAI: Deterministic Inference, Verifiable Results. *arXiv preprint arXiv:2602.00182*, 2026. URL https://arxiv.org/abs/2602.00182.
4. Yvo Desmedt and Yair Frankel. Threshold cryptosystems. In *Advances in Cryptology – CRYPTO '89 Proceedings*, Lecture Notes in Computer Science, pages 307–315. Springer, 1990. doi: 10.1007/0-387-34805-0_28.
5. Paul Feldman. A practical scheme for non-interactive verifiable secret sharing. In *28th Annual Symposium on Foundations of Computer Science (FOCS)*, pages 427–438, 1987. doi: 10.1109/SFCS.1987.4.
6. Berry Schoenmakers. A simple publicly verifiable secret sharing scheme and its application to electronic voting. In *CRYPTO*, pages 148–164, 1999. doi: 10.1007/3-540-48405-1_10.
7. Amir Herzberg, Stanislaw Jarecki, Hugo Krawczyk, and Moti Yung. Proactive secret sharing or: How to cope with perpetual leakage. In *CRYPTO*, pages 339–352, 1995.
8. Ari Juels and Burton S. Kaliski, Jr. PORs: Proofs of retrievability for large files. In *Proceedings of the 14th ACM Conference on Computer and Communications Security (CCS '07)*, pages 584–597. ACM, 2007. doi: 10.1145/1315245.1315317.
9. Mustafa Al-Bassam, Alberto Sonnino, Vitalik Buterin, and Ismail Khoffi. Fraud and data availability proofs: Detecting invalid blocks in light clients. In *Financial Cryptography and Data Security (FC 2021)*, Lecture Notes in Computer Science, pages 279–298. Springer, 2021. doi: 10.1007/978-3-662-64331-0_15.
10. Christian Cachin, Klaus Kursawe, and Victor Shoup. Random oracles in Constantinople: Practical asynchronous Byzantine agreement using cryptography. *Journal of Cryptology*, 18(3):219–246, 2005. doi: 10.1007/s00145-005-0318-0.
11. Alexandra Boldyreva. Threshold signatures, multisignatures and blind signatures based on the gap-Diffie-Hellman-group signature scheme. In *Public Key Cryptography – PKC 2003*, Lecture Notes in Computer Science, pages 31–46. Springer, 2003. doi: 10.1007/3-540-36288-6_3.
12. Henry de Valence, Jack Grigg, Mike Hamburg, Isis Lovecruft, George Tankersley, and Filippo Valsorda. The ristretto255 and decaf448 groups. Technical Report RFC 9496, Internet Engineering Task Force, 2023. URL https://www.rfc-editor.org/rfc/rfc9496.
13. Daniel Kahn Gillmor. Negotiated finite field Diffie-Hellman ephemeral parameters for transport layer security (TLS). Request for Comments 7919, Internet Engineering Task Force, 2016. URL https://www.rfc-editor.org/rfc/rfc7919.
14. Adi Shamir. How to share a secret. *Communications of the ACM*, 22(11):612–613, 1979.
15. Victor Shoup and Rosario Gennaro. Securing threshold cryptosystems against chosen ciphertext attack. *Journal of Cryptology*, 15(2):75–96, 2002. doi: 10.1007/s00145-001-0020-9.
16. David Chaum and Torben Pryds Pedersen. Wallet databases with observers. In *Advances in Cryptology – CRYPTO '92*, pages 89–105. Springer, 1993. doi: 10.1007/3-540-48071-4_7.
17. Amos Fiat and Adi Shamir. How to prove yourself: Practical solutions to identification and signature problems. In *Advances in Cryptology – CRYPTO '86*, Lecture Notes in Computer Science, pages

186–194. Springer, 1987. doi: 10.1007/3-540-47721-7_12.

18. Sourav Das, Vinith Krishnan, Irene Miriam Isaac, and Ling Ren. SPURT: Scalable distributed randomness beacon with transparent setup. In *2022 IEEE Symposium on Security and Privacy (SP)*, pages 2502–2517. IEEE, 2022. doi: 10.1109/SP46214.2022.9833580.
19. Sai Krishna Deepak Maram, Fan Zhang, Lun Wang, Andrew Low, Yupeng Zhang, Ari Juels, and Dawn Song. CHURP: Dynamic-committee proactive secret sharing. In *Proceedings of the 2019 ACM SIGSAC Conference on Computer and Communications Security (CCS '19)*, pages 2369–2386. ACM, 2019. doi: 10.1145/3319535.3363203.
20. Thomas Yurek, Zhuolun Xiang, Yu Xia, and Andrew Miller. Long live the honey badger: Robust asynchronous DPSS and its applications. In *32nd USENIX Security Symposium (USENIX Security 23)*, pages 5413–5430. USENIX Association, 2023. URL `https://www.usenix.org/conference/usenixsecurity23/presentation/yurek`.
21. Rosario Gennaro, Stanisław Jarecki, Hugo Krawczyk, and Tal Rabin. Secure distributed key generation for discrete-log based cryptosystems. *Journal of Cryptology*, 20(1):51–83, 2007. doi: 10.1007/s00145-006-0347-3.
22. Sourav Das, Thomas Yurek, Zhuolun Xiang, Andrew Miller, Lefteris Kokoris-Kogias, and Ling Ren. Practical asynchronous distributed key generation. In *2022 IEEE Symposium on Security and Privacy (SP)*, pages 2518–2534. IEEE, 2022. doi: 10.1109/SP46214.2022.9833584.
23. Armando Faz-Hernandez, Sam Scott, Nick Sullivan, Riad S. Wahby, and Christopher A. Wood. Hashing to elliptic curves. Technical Report RFC 9380, Internet Engineering Task Force, 2023. URL `https://www.rfc-editor.org/rfc/rfc9380`.
24. Dan Boneh, Benedikt Bünz, Kartik Nayak, Lior Rotem, and Victor Shoup. Context-dependent threshold decryption and its applications. Cryptology ePrint Archive, Paper 2025/279; extended abstract in ASIACRYPT 2025, 2025. URL `https://eprint.iacr.org/2025/279`.
25. Abraham Wald. Sequential tests of statistical hypotheses. *The Annals of Mathematical Statistics*, 16(2):117–186, 1945. doi: 10.1214/aoms/1177731118.
26. Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963. doi: 10.1080/01621459.1963.10500830.
27. Edwin B. Wilson. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927. doi: 10.1080/01621459.1927.10502953.
28. Daniel Ford, François Labelle, Florentina I. Popovici, Murray Stokely, Van-Anh Truong, Luiz Barroso, Carrie Grimes, and Sean Quinlan. Availability in globally distributed storage systems. In *9th USENIX Symposium on Operating Systems Design and Implementation (OSDI 10)*. USENIX Association, 2010. URL `https://www.usenix.org/conference/osdi10/availability-globally-distributed-storage-systems`.
29. Andrew Lewis-Pye, Joachim Neu, Tim Roughgarden, and Luca Zanolini. Accountable liveness. *arXiv preprint arXiv:2504.12218*, 2025. URL `https://arxiv.org/abs/2504.12218`.
30. Eleftherios Kokoris-Kogias, Enis Ceyhun Alp, Linus Gasser, Philipp Jovanovic, Ewa Syta, and Bryan Ford. CALYPSO: Private data management for decentralized ledgers. *Proceedings of the VLDB Endowment*, 14(4):586–599, 2020. doi: 10.14778/3436905.3436917.
31. Oliver Stengele, Markus Raiber, Jörn Müller-Quade, and Hannes Hartenstein. ETHTID: Deployable threshold information disclosure on Ethereum. In *2021 3rd International Conference on Blockchain Computing and Applications (BCCA)*, pages 127–134, 2021. doi: 10.1109/BCCA53669.2021.9657019.

URL https://arxiv.org/abs/2107.01600.

32. Joseph Bebel and Dev Ojha. Ferveo: Threshold decryption for mempool privacy in BFT networks. Cryptology ePrint Archive, Paper 2022/898, 2022. URL https://eprint.iacr.org/2022/898.
33. Jan Bormet, Sebastian Faust, Hussien Othman, and Ziyan Qu. BEAT-MEV: Epochless approach to batched threshold encryption for MEV prevention. In *34th USENIX Security Symposium (USENIX Security 25)*. USENIX Association, 2025. URL https://www.usenix.org/conference/usenixsecurity25/presentation/bormet.
34. Arka Rai Choudhuri, Sanjam Garg, Julien Piet, and Guru-Vamsi Policharla. Mempool privacy via batched threshold encryption: Attacks and defenses. In *33rd USENIX Security Symposium (USENIX Security 24)*, pages 3513–3529. USENIX Association, 2024. ISBN 978-1-939133-44-1. URL https://www.usenix.org/conference/usenixsecurity24/presentation/choudhuri.
35. Peyman Momeni, Sergey Gorbunov, and Bohan Zhang. FairBlock: Preventing blockchain front-running with minimal overheads. In *Security and Privacy in Communication Networks (SecureComm 2022)*, Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering, pages 250–271. Springer, 2023. doi: 10.1007/978-3-031-25538-0_14.
36. Haoqian Zhang, Louis-Henri Merino, Ziyan Qu, Mahsa Bastankhah, Vero Estrada-Galiñanes, and Bryan Ford. F3B: A low-overhead blockchain architecture with per-transaction front-running protection. In *5th Conference on Advances in Financial Technologies (AFT 2023)*, volume 282 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 3:1–3:23. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2023. doi: 10.4230/LIPIcs.AFT.2023.3.
37. Jingyu Liu, Jingjing Rao, and Tetsutaro Uehara. UQ-Vote: Consensus-authorized threshold decryption for end-to-end verifiable elections. *IEEE Access*, 14:128905–128916, 2026. doi: 10.1109/ACCESS.2026.3725080.
38. Giuseppe Ateniese, Randal Burns, Reza Curtmola, Joseph Herring, Lea Kissner, Zachary Peterson, and Dawn Song. Provable data possession at untrusted stores. In *Proceedings of the 14th ACM Conference on Computer and Communications Security (CCS '07)*, pages 598–609. ACM, 2007. doi: 10.1145/1315245.1315318.
39. Kevin D. Bowers, Ari Juels, and Alina Oprea. HAIL: A high-availability and integrity layer for cloud storage. In *Proceedings of the 16th ACM Conference on Computer and Communications Security (CCS '09)*, pages 187–198. ACM, 2009. doi: 10.1145/1653662.1653686.
40. Dankrad Feist. Proofs of custody. Blog post, 2021. URL https://dankradfeist.de/ethereum/2021/09/30/proofs-of-custody.html. Accessed 14 October 2026.
41. Threshold Network. tBTC v2 bridge: Heartbeat and moving funds. Developer documentation, 2026. URL https://docs.threshold.network/app-development/tbtc-v2/tbtc-contracts-api/tbtc-v2-api/heartbeat. Accessed 14 October 2026.
42. Mathias Hall-Andersen, Mark Simkin, and Benedikt Wagner. Foundations of data availability sampling. *IACR Communications in Cryptology*, 1(4), 2025. doi: 10.62056/a09qudhdj.
43. Vitalik Buterin and Virgil Griffith. Casper the friendly finality gadget. arXiv preprint arXiv:1710.09437, 2017. URL https://arxiv.org/abs/1710.09437.
44. Peiyao Sheng, Gerui Wang, Kartik Nayak, Sreeram Kannan, and Pramod Viswanath. BFT protocol forensics. In *Proceedings of the 2021 ACM SIGSAC Conference on Computer and Communications Security (CCS '21)*, pages 1722–1743. ACM, 2021. doi: 10.1145/3460120.3484566.

45. Dan Boneh, Aditi Partap, and Lior Rotem. Accountability for misbehavior in threshold decryption via threshold traitor tracing. In *Advances in Cryptology – CRYPTO 2024*, Lecture Notes in Computer Science, pages 317–351. Springer, 2024. doi: 10.1007/978-3-031-68394-7_11.